UNIVERSITY OF LONDON

BSc EXAMINATION 2023

For Internal Students of
Royal Holloway

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IY3609: DIGITAL FORENSICS

IY3609R: DIGITAL FORENSICS – PAPER FOR FIRST SITS/RESIT CANDIDATES

Time Allowed: **TWO** hours

Attempt **three** questions.
No credit will be given for attempting any further questions.

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Question 1 Legal Matters

(a) Laura is a digital forensics expert who appears as an expert witness in a criminal case. She presents a novel analysis method for scenarios that involve cloud services, which led to finding a specific piece of evidence.

Describe three typical questions that Laura might have to answer regarding her novel analysis method to justify her finding. For this, argue in light of the *Daubert* criteria, which establish a standard for the quality of expert testimonies that is practically relevant to UK courts. How could she respond to these questions? [13]

(b) Ralph is a UK national who is using an Internet service provider that is located in Spain. Ralph is accused of criminal VAT evasion in the UK.

Assume the responsible UK court issues an Overseas Production Order (OPO) demanding electronic evidence from the Spanish provider for the purpose of the criminal investigation.

Is the Spanish provider obligated to provide the requested evidence, and which response can the UK court expect? Give a reasoned answer. [12]

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Question 2 Storage Forensics

Carl is a forensic investigator who is supposed to analyze an external hard disk that was seized during a police raid.

(a) Unfortunately, Carl has not received any information about the disk's contents or which computer it has been connected to.

Describe how Carl can determine which file systems are present on the disk. Additionally, describe how can Carl make sure that there is no unused space on the disk that could contain hidden information.

Assume that the disk as well as its logical structure are undamaged; clearly state further assumptions that you make. [8]

(b) As part of his preliminary investigation under 2(a), Carl finds a **damaged** FAT file system. Specifically, the beginning of the file system structure (including root directory and the file allocation table as well as its backup structure) have been overwritten.

(i) Describe in detail how Carl can recover the remaining logical structure of the damaged FAT file system (i.e., directories and files, including fragments thereof) and limitations of that endeavor. [8]

(ii) Carl is supposed to look out specifically for JPEG pictures. Describe how Carl can determine whether files of such type are present in the damaged FAT file system. Additionally, describe the issues he might face when trying to recover the content of files. [6]

(iii) Briefly describe under which conditions Carl will be able to recover metadata for files in the damaged FAT file system. [3]

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Question 3 Memory Forensics

Helen is a forensic investigator examining a laptop computer that is running Microsoft Windows 10 and that is part of an Active Directory (AD) domain. She suspects that the machine is compromised by a rootkit.

(a) The machine is currently running and connected to the network.

(i) How can Helen take a memory image of the machine? Assume that no memory snapshot agent is installed and secure boot is disabled.

Propose one technique. Detail how the image is taken and any limitations of the proposed technique. State all additional assumptions you make. [5]

(ii) Can the rootkit detect and counteract the taking of a memory image when using the technique described in the answer to question 3(a)(i)?

Give a reasoned answer and detail how effective potential countermeasures will be. [10]

(b) Assume that Helen was successful in capturing the memory of the live system (cf. question 3(a)(i)). Based on the memory image, she now tries to figure out whether the rootkit potentially has compromised any AD accounts on the machine.

Make a recommendation and explain which structures she should investigate to come to a conclusion. [10]

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Question 4 Mobile Forensics

Daniel analyzes a seized Apple iPhone 8 with iOS 16 installed whose owner recently deceased.

(a) The phone is powered on, but the screen is locked.

(i) The device is equipped with a fingerprint scanner for authentication purposes.

Describe how Daniel can potentially unlock the phone without manipulating its memory. Also describe potential limitations of the proposed approach. [6]

(ii) Usually, mobile phones offer fallback mechanisms in case biometric authentication fails, e.g., they require to enter a PIN.

Describe common safeguards that prevent Daniel from efficiently brute-forcing the PIN. [6]

(b) Daniel could also try to exploit vulnerabilities of the device's operating system (as commonly done during "jailbreaking") to bypass the authentication mechanisms.

(i) Assume that the chosen bypassing method requires to re-start the device.

Discuss risks attached to this method that might result in irretrievably losing access to the desired data. [6]

(ii) Assume that Daniel successfully bypasses the authentication mechanism using a "jailbreaking" tool that he found on the Internet.

Why might conclusions drawn from a memory image obtained this way not be accepted by a UK court in a criminal case? Give a reasoned answer. [7]

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Question 5 Network Forensics

Hannah is analyzing a host system that is suspected to be infected with malware. Since she is not supposed to interfere with the system in any way, she is restricted to analyze its network traffic.

(a) To confirm the hypothesis that the system is infected, Hannah is determined to find evidence that the malware frequently contacts so-called "command and control" (C2) servers. This is difficult since malware will often use obfuscated and encrypted channels.

(i) Hannah starts her analysis by looking for suspicious network traffic patterns. Describe three possible techniques that the malware could use to obfuscate its network traffic **patterns** (i.e., beyond using encryption). [6]

(ii) Describe two methods how Hannah can nevertheless identify network traffic patterns that indicate communication with a C2 server. [6]

(b) Unfortunately, Hannah was not successful in capturing network traffic of an interaction with a C2 server during her analysis (cf. question 5(a)). However, assume that some communication of this kind has indeed taken place.

How could the malware potentially evade detection? Formulate and justify a hypothesis. How could this hypothesis be confirmed? [8]

(c) Due to the lack of success (cf. question 5(b)), Hannah is given permission to attempt memory analysis instead. For this, she captures a memory image of the host soon after the hypothesised communication had taken place.

Discuss which information Hannah might be able to infer from inspecting the connection table structure for TCP connections in the memory image. [5]

END

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